Linear Transformations

15 October 2025

Study Guide

This week, we will be learning about:

- linear transformations;
- geometric transformations in $\mathbb{R}^2$: scaling, projections, reflections, rotations, and shears;
- the inverse of a linear transformation.

You may find it helpful to list down many examples *and* counterexamples of the various objects we encounter in this week's class. Moreover, we talk about the following key results:

Theorem 1. *Let $T : \mathbb{R}^m \to \mathbb{R}^n$ be a linear transformation. Then, the standard matrix of T is given by*

$$\boldsymbol{A} = \begin{pmatrix} \uparrow & \uparrow & & \uparrow \\ T(\vec{e}_1) & T(\vec{e}_2) & \cdots & T(\vec{e}_m) \\ \downarrow & \downarrow & & \downarrow \end{pmatrix},$$

where $\vec{e}_i$ is the ith standard basis vector in $\mathbb{R}^m$.

Theorem 2. *Let $T : \mathbb{R}^m \to \mathbb{R}^n$ be a function. Then, T is a linear transformation if and only if, for all $\vec{v}, \vec{w} \in \mathbb{R}^m$:*

1. *$T(\vec{v} + \vec{w}) = T(\vec{v}) + T(\vec{w})$, and*
2. *$T(k\vec{v}) = kT(\vec{v})$ for all $k \in \mathbb{R}$.*

Theorem 3. *Let $\boldsymbol{A}$ be an $n \times n$ matrix. Then, $\boldsymbol{A}$ is invertible if and only if the reduced row-echelon form of $\boldsymbol{A}$ is the identity matrix I_n.*

Theorem 4 (Equivalent Statements)**.** *Let $\boldsymbol{A}$ be an $n \times n$ matrix. The following statements are equivalent:*

1. *$\boldsymbol{A}$ is invertible; that is, $\boldsymbol{A}^{-1}$ exists.*
2. *The reduced row echelon form of $\boldsymbol{A}$ is $\boldsymbol{I}_n$.*
3. *$\boldsymbol{Ax} = \boldsymbol{b}$ has a unique solution for each $n \times 1$ matrix $\boldsymbol{b}$, namely $\boldsymbol{x} = \boldsymbol{A}^{-1}\boldsymbol{b}$.*
4. *$\boldsymbol{Ax} = \boldsymbol{0}$ has only the trivial solution.*

Definitions

Definition 5. Let X and Y be two sets. A **function** $T : X \to Y$ is a rule that associates each element of the **domain** X to a unique element in the **codomain** Y. A function $T : \mathbb{R}^m \to \mathbb{R}^n$ is a **linear transformation** if there exists an $n \times m$ matrix $\boldsymbol{A}$ such that

$$T(\vec{x}) = \boldsymbol{A}\vec{x} \text{ for all } \vec{x} \in \mathbb{R}^m.$$

The matrix $\boldsymbol{A}$ is called the **standard matrix** of the transformation T.

Definition 6. A function $T : X \to Y$ is said to be **invertible** if the equation $T(x) = y$ has a unique solution $x \in X$ for every $y \in Y$, and we write

$$T^{-1}(y) = x.$$

A square matrix $\boldsymbol{A}$ is **invertible** if the linear transformation $T(\vec{x}) = \boldsymbol{A}\vec{x}$ is invertible, and we let $\boldsymbol{A}^{-1}$ denote the standard matrix of T^{-1}.

Exercises

1. Let

$$\vec{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{w} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

   On the same set of axes, sketch the images of $\vec{v}$ and $\vec{w}$ under the linear transformation

$$T(\vec{x}) = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix} \vec{x}.$$

2. Determine the standard matrix of the orthogonal projection onto the line ℓ given by

$$\ell = t \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \quad t \in \mathbb{R}.$$

3. Consider the following matrices:

$$\boldsymbol{A} = \begin{pmatrix} 0.6 & 0.8 \\ 0.8 & -0.6 \end{pmatrix}, \quad \boldsymbol{B} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}, \quad \boldsymbol{C} = \begin{pmatrix} 0.36 & -0.48 \\ -0.48 & 0.64 \end{pmatrix},$$

$$\boldsymbol{D} = \begin{pmatrix} -0.8 & 0.6 \\ -0.6 & -0.8 \end{pmatrix}, \quad \boldsymbol{E} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}.$$

   What geometric transformation does each matrix correspond to?

4. Let the 2×2 matrices $\boldsymbol{A}$ be given by

$$\boldsymbol{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

   Show that if $ad - bc \neq 0$, then $\boldsymbol{A}$ is invertible and

$$\boldsymbol{A}^{-1} = \begin{pmatrix} \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}.$$

5. Let $\boldsymbol{M}$ be the matrix given by

$$\boldsymbol{M} = \begin{pmatrix} 0 & 1 & 3 & -1 \\ 2 & 0 & 4 & 0 \\ -2 & 1 & 0 & -2 \\ 0 & 0 & 2 & 1 \end{pmatrix}.$$

   Determine whether or not $\boldsymbol{M}$ is invertible, and find $\boldsymbol{M}^{-1}$ if it exists.